Harsh Hirawat

AI Engineer — LLM Systems & Applied AI

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Education

Newton School of Technology (ADYPU) — Pune, Maharashtra, India
B.Tech in Computer Science & Engineering **(CGPA: 8.86 / 10.0)** — *Aug 2024 – May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems (DBMS), Distributed Systems.

Technical Skills

AI & LLM Engineering: LangChain, LangGraph, LLM APIs (OpenAI, Gemini Live), Corrective RAG (CRAG), Pinecone, pgvector, Vector Search, Semantic Reranking, Embeddings, Prompt Engineering
Programming Languages: TypeScript (Primary), JavaScript, Python (Secondary), SQL, Bash, C / C++ (Foundational)
Machine Learning & Evaluation: PyTorch, Retrieval Benchmarking, Groundedness & Faithfulness Evaluation
Backend & Data Systems: FastAPI, Node.js, Fastify, Express, PostgreSQL (Prisma, pooling), Redis, MongoDB, BullMQ, WebSockets
Cloud, Systems & Observability: Docker, Kubernetes, AWS, Cloudflare, CI/CD, OpenTelemetry, Prometheus, Grafana, WebRTC, Linux

Work Experience

AI Engineer Intern — Oct 2025 – March 2026
CommKraft — *Remote*

- Engineered a real-time voice AI system integrating multimodal LLM APIs, WebRTC streaming, and asynchronous event pipelines, supporting 200+ concurrent user sessions.
- Optimized LLM orchestration and multi-tier caching, reducing end-to-end response latency from ∼3s to <300ms.
- Implemented a state-driven conversation engine with interruption handling, tool execution, and automated fallback workflows, achieving sub-500ms failure recovery.

Freelance Full-Stack & Systems Developer — Oct 2025 – Present
MindReader Biotech — *USA (Remote)*

- Architected containerized microservices platform with in-memory caching, reducing deployment turnaround by ∼50% via automated CI/CD pipelines.
- Engineered cloud edge infrastructure with automated failover and dynamic traffic routing, sustaining ∼99.9% uptime across production workloads.
- Built 5+ high-performance analytical dashboards with responsive server-side data workflows, reducing data processing latency by ∼35%.

Projects

AskMyNotes | *Citation-Backed Corrective RAG & Real-Time Search* | askmynotes.greenhacker.in | GitHub

- Built a corrective RAG (CRAG) pipeline for document question answering using managed vector databases, PostgreSQL, dense semantic embeddings, and state-graph nodes for multi-turn conversational memory.
- Implemented subject-scoped semantic indexing, hybrid reranking (75% semantic + 25% lexical), and confidence-threshold gating, improving retrieval precision by ∼30% over baseline keyword search.
- Developed a low-latency streaming chat interface via WebSockets with per-user document isolation, vector caching, and full-duplex conversational voice interaction.

Campus Connect | *Distributed Mentorship Marketplace & Real-Time Platform* | campus-connect.co.in | GitHub

- Architected a distributed web platform serving **14.6M+ requests** and **230+ GB bandwidth**, absorbing viral traffic surges of **30k+ visits/day** across 6+ countries via global edge caching.
- Implemented **Blue/Green zero-downtime deployment pipelines** with active health-check probes and automated traffic steering, sustaining **120+ days continuous uptime (99.99% availability)** with zero-latency rollbacks.
- Engineered real-time WebRTC media streaming and asynchronous distributed task queues to decouple booking holds, payments, and notifications, eliminating database race conditions.

Evergreen Classic | *E-Commerce Platform & Conversational AI Engine* | evergreenclassic.in | WhatsApp

- Built an asynchronous e-commerce platform integrating conversational AI, vector embeddings, and PostgreSQL connection pooling for semantic catalog search.
- Engineered real-time webhook ingestion and background worker queues for multi-channel customer interactions (WhatsApp and Web) with automated order processing.

Honors, Achievements & Certifications

Hackathons: Finalist, Mumbai Hacks | **Semi-Finalist, Google GenAI Hackathon**
Open Source & Problem Solving: Active contributor with **500+ GitHub commits** across distributed systems and AI repositories; solved algorithmic problems in Python and TypeScript on LeetCode & HackerRank
Certifications: AI Engineer Associate Certificate | AI Hero: 12-Month Journey from Zero to Expert (Udemy)